

Table 1: End-to-end execution time (s, median per iteration) and speedup split by pass type. ADT fields = non-recursive fields annotated with @BENCH adt_fields; SoA bufs = 1+ non-recursive field slots across all constructors (recursive children are stored in the tag buffer). Speedup >1× means SoA is faster; **bold** marks >1.1×.

Program	ADT fields	SoA bufs	Fold passes			Map passes		
			AoS (s)	SoA (s)	Speedup	AoS (s)	SoA (s)	Speedup
Compiler	9	8	0.1235	0.0388	3.18 ×	0.1585	0.1895	0.84×
DBQuery	15	13	0.0626	0.0384	1.63 ×	0.1736	0.2482	0.70×
DecisionTree	7	6	3.722	2.989	1.25 ×	—	—	—
DomTree	15	14	0.1341	0.0617	2.17 ×	0.0585	0.0895	0.65×
KDTree	19	12	0.3661	0.2895	1.26 ×	—	—	—
LinearListReduction	11	11	0.0274	0.0049	5.59 ×	—	—	—
List	2	2	0.0876	0.0906	0.97×	0.2943	0.3328	0.88×
MonoTree	3	2	0.0236	0.0225	1.05×	0.0462	0.0421	1.10×
ObjectGraph	5	4	0.0066	0.0051	1.29 ×	0.0184	0.0200	0.92×
OctTree	16	9	0.0158	0.0130	1.21 ×	0.0384	0.0491	0.78×
PiecewiseFunctions	8	7	0.0102	0.0055	1.85 ×	0.0450	0.0511	0.88×
TernaryTree	5	3	0.0301	0.0283	1.06×	0.0845	0.0771	1.10×
Trie	10	9	0.0246	0.0111	2.21 ×	0.0755	0.0945	0.80×

Table 2: PAPI speedup summary across all passes. For each program and counter, speedup is AoS/SoA using summed per-pass median counters. >1× means SoA has fewer events.

Program	CYC	INS	L1D	L1I	L2D	L2I	LLC
Compiler	1.43 ×	0.48×	4.02 ×	0.66×	15.75 ×	0.39×	6.94 ×
DBQuery	0.74×	0.37×	2.23 ×	0.47×	1.33 ×	0.12×	6.10 ×
DecisionTree	1.25 ×	0.69×	18.82 ×	2.78 ×	0.82×	2.92 ×	36.57 ×
DomTree	1.36 ×	0.53×	2.72 ×	1.03×	21.51 ×	0.36×	10.90 ×
KDTree	1.27 ×	0.87×	6.32 ×	2.53 ×	0.32×	2.50 ×	3.48 ×
LinearListReduction	5.62 ×	0.90×	80.79 ×	15.91 ×	264.05 ×	15.62 ×	11.39 ×
List	0.89×	0.74×	3.54 ×	0.49×	0.12×	0.59×	1.44 ×
MonoTree	1.12 ×	0.82×	0.32×	0.66×	0.23×	0.74×	1.13 ×
ObjectGraph	1.00×	0.69×	2.93 ×	0.94×	0.59×	2.01 ×	4.50 ×
OctTree	0.85×	0.55×	0.34×	0.47×	0.31×	0.72×	6.12 ×
PiecewiseFunctions	0.96×	0.52×	2.28 ×	0.73×	1.15 ×	0.52×	6.08 ×
TernaryTree	1.16 ×	0.78×	0.35×	0.52×	0.25×	1.04×	1.35 ×
Trie	0.91×	0.47×	3.13 ×	0.71×	2.79 ×	0.11×	8.81 ×
Geomean	1.20 ×	0.62×	2.81 ×	1.02×	1.45 ×	0.81×	5.23 ×

Table 3: PAPI counter totals across all passes (AoS/SoA). Each entry is the sum of per-pass median counter values.

Program	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
Compiler	9.3e+08/ 6.5e+08	9.9e+08 /2.1e+09	1.2e+07/ 3.0e+06	2.4e+05 /3.6e+05	1.3e+06/ 8.0e+04	1.5e+04 /3.9e+04	5.1e+07/ 7.4e+06
DBQuery	6.9e+08 /9.3e+08	1.3e+09 /3.5e+09	1.0e+07/ 4.7e+06	2.6e+05 /5.6e+05	9.7e+04/ 7.3e+04	1.1e+04 /9.5e+04	2.6e+07/ 4.2e+06
DecisionTree	1.9e+10/ 1.5e+10	4.3e+10 /6.3e+10	5.9e+08/ 3.1e+07	1.6e+05/ 5.9e+04	1.7e+06 /2.1e+06	1.6e+05/ 5.6e+04	1.1e+09/ 3.0e+07
DomTree	8.1e+08/ 5.9e+08	1.2e+09 /2.3e+09	1.1e+07/ 4.1e+06	1.2e+05/ 1.2e+05	1.1e+06/ 4.9e+04	1.2e+04 /3.3e+04	5.7e+07/ 5.2e+06
KDTree	1.9e+09/ 1.5e+09	6.9e+09 /8.0e+09	2.2e+07/ 3.5e+06	1.2e+03/ 4.7e+02	4.3e+04 /1.3e+05	1.2e+03/ 4.7e+02	8.1e+07/ 2.3e+07
LinearListReduction	1.4e+08/ 2.5e+07	9.0e+07 /1.0e+08	4.7e+06/ 5.8e+04	7.3e+02/ 4.6e+01	1.9e+06/ 7.2e+03	7.3e+02/ 4.7e+01	1.2e+07/ 1.1e+06
List	1.2e+09 /1.4e+09	4.5e+09 /6.1e+09	9.7e+06/ 2.7e+06	3.0e+05 /6.1e+05	2.2e+04 /1.8e+05	1.1e+04 /1.9e+04	4.8e+07/ 3.4e+07
MonoTree	2.9e+08/ 2.6e+08	9.1e+08 /1.1e+09	1.7e+05 /5.3e+05	5.3e+04 /8.1e+04	3.3e+03 /1.5e+04	4.5e+03 /6.1e+03	1.9e+06/ 1.7e+06
ObjectGraph	8.3e+07 /8.4e+07	2.6e+08 /3.8e+08	3.7e+05/ 1.3e+05	2.7e+04 /2.8e+04	2.7e+03 /4.6e+03	2.9e+03/ 1.5e+03	5.3e+05/ 1.2e+05
OctTree	1.7e+08 /2.0e+08	4.1e+08 /7.5e+08	3.3e+05 /9.6e+05	5.3e+04 /1.1e+05	4.9e+03 /1.6e+04	2.9e+03 /4.1e+03	3.6e+06/ 5.8e+05
PiecewiseFunctions	1.6e+08 /1.6e+08	3.5e+08 /6.6e+08	1.7e+06/ 7.5e+05	6.0e+04 /8.2e+04	1.4e+04/ 1.2e+04	5.6e+03 /1.1e+04	3.3e+06/ 5.4e+05
TernaryTree	4.3e+08/ 3.7e+08	1.2e+09 /1.6e+09	4.0e+05 /1.1e+06	7.2e+04 /1.4e+05	5.1e+03 /2.0e+04	4.0e+03/ 3.9e+03	3.2e+06/ 2.4e+06
Trie	2.9e+08 /3.2e+08	5.2e+08 /1.1e+09	3.9e+06/ 1.2e+06	1.1e+05 /1.6e+05	6.8e+04/ 2.5e+04	3.2e+03 /3.0e+04	1.0e+07/ 1.2e+06

Table 4: Mutable vs immutable cursor comparison (times only). Times are median per iteration (s). **Bold** marks the fastest time across all four variants.

Program	Am	Ai	Sm	Si
Compiler	0.2819	1.194	0.2283	<i>OOM</i>
DBQuery	0.2362	0.3166	0.2866	1.378
DecisionTree	3.722	5.674	2.989	28.40
DomTree	0.1926	0.3376	0.1512	1.023
KDTree	0.3661	0.4402	0.2895	2.626
LinearListReduction	0.0274	0.1652	0.0049	<i>OOM</i>
List	0.3820	<i>OOM</i>	0.4234	<i>OOM</i>
MonoTree	0.0698	0.1490	0.0645	0.2941
ObjectGraph	0.0250	0.0445	0.0251	0.0784
OctTree	0.0542	0.0597	0.0622	0.2471
PiecewiseFunctions	0.0552	0.0750	0.0566	0.2939
TernaryTree	0.1145	0.1492	0.1054	0.6177
Trie	0.1000	0.1354	0.1056	0.4692

Table 5: Mutable vs immutable cursor comparison (speedups). Shown: AoS-mut/SoA-mut, AoS-imm/AoS-mut, AoS-imm/SoA-mut, AoS-imm/SoA-imm. >1× means the denominator is faster.

Program	Am/Sm	Ai/Am	Ai/Sm	Ai/Si
Compiler	1.24 ×	4.24 ×	5.23 ×	—
DBQuery	0.82×	1.34 ×	1.10 ×	0.23×
DecisionTree	1.25 ×	1.52 ×	1.90 ×	0.20×
DomTree	1.27 ×	1.75 ×	2.23 ×	0.33×
KDTree	1.26 ×	1.20 ×	1.52 ×	0.17×
LinearListReduction	5.59 ×	6.03 ×	33.68 ×	—
List	0.90×	—	—	—
MonoTree	1.08×	2.14 ×	2.31 ×	0.51×
ObjectGraph	1.00×	1.78 ×	1.77 ×	0.57×
OctTree	0.87×	1.10 ×	0.96×	0.24×
PiecewiseFunctions	0.98×	1.36 ×	1.33 ×	0.26×
TernaryTree	1.09×	1.30 ×	1.42 ×	0.24×
Trie	0.95×	1.35 ×	1.28 ×	0.29×

Table 6: Per-pass performance for **Compiler**, ADT has 9 fields, SoA uses 8 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across –iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Si	Am/Sm	Ai/Am	Ai/Sm	Ai/Si
blockCountPass	F	2/9	78%	0.0116 $\pm$ 0.00000	0.0845 $\pm$ 0.00003	0.0019$\pm$0.00000	OOM	6.03$\times$	7.29$\times$	44.01$\times$	–
branchStatsPass	F	2/9	78%	0.0113 $\pm$ 0.00000	0.0964 $\pm$ 0.00002	0.0034$\pm$0.00000	OOM	3.31$\times$	8.56$\times$	28.37$\times$	–
castInstCountPass	F	3/9	67%	0.0177 $\pm$ 0.00000	0.0956 $\pm$ 0.00100	0.0013$\pm$0.00000	OOM	13.74$\times$	5.40$\times$	74.25$\times$	–
has cycle	F	4/9	56%	0.0249 $\pm$ 0.00015	0.0973 $\pm$ 0.00002	0.0171$\pm$0.00015	OOM	1.45$\times$	3.91$\times$	5.68$\times$	–
instCountPass	F	2/9	78%	0.0116 $\pm$ 0.00001	0.0842 $\pm$ 0.00103	0.0019$\pm$0.00000	OOM	6.05$\times$	7.25$\times$	43.86$\times$	–
latencyModelPass	F	3/9	67%	0.0115 $\pm$ 0.00000	0.0961 $\pm$ 0.00017	0.0030$\pm$0.00000	OOM	3.87$\times$	8.35$\times$	32.31$\times$	–
memoryOpStatsPass	F	3/9	67%	0.0114 $\pm$ 0.00000	0.0955 $\pm$ 0.00003	0.0038$\pm$0.00000	OOM	2.99$\times$	8.38$\times$	25.06$\times$	–
stripSideEffectsPass	M	7/9	22%	0.0799$\pm$0.00082	0.1918 $\pm$ 0.00022	0.0963 $\pm$ 0.00112	OOM	0.83 $\times$	2.40$\times$	1.99$\times$	–
targetReturnPass	M	9/9	0%	0.0786$\pm$0.00017	0.1877 $\pm$ 0.00108	0.0932 $\pm$ 0.00013	OOM	0.84 $\times$	2.39$\times$	2.01$\times$	–
throughputModelPass	F	3/9	67%	0.0115 $\pm$ 0.00000	0.0957 $\pm$ 0.00016	0.0030$\pm$0.00000	OOM	3.91$\times$	8.29$\times$	32.41$\times$	–
verifyIR	F	9/9	0%	0.0120 $\pm$ 0.00015	0.0694 $\pm$ 0.00116	0.0034$\pm$0.00005	OOM	3.50$\times$	5.77$\times$	20.23$\times$	–
Total				0.2819	1.194	0.2283	–	1.24$\times$	4.24$\times$	5.23$\times$	–
Geomean								3.11$\times$	5.64$\times$	17.57$\times$	–

Table 7: Per-pass PAPI counters for **Compiler** (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
blockCountPass	F	5.9e+07/ 9.8e+06	5.1e+07/ 5.1e+07	1.3e+06/ 1.9e+04	5.8e+02/ 2.2e+01	2.2e+05/ 1.1e+03	5.8e+02/ 2.3e+01	4.7e+06/ 2.2e+03
branchStatsPass	F	5.7e+07/ 1.7e+07	7.9e+07 /8.6e+07	8.4e+05/ 2.4e+04	5.0e+02/ 5.5e+01	4.8e+04/ 3.4e+03	5.0e+02/ 5.5e+01	4.6e+06/ 4.4e+05
castInstCountPass	F	9.0e+07/ 6.5e+06	5.1e+07/ 3.2e+07	1.3e+04 /1.7e+04	5.6e+01/ 2.6e+01	7.8e+02 /1.1e+03	5.9e+01/ 2.7e+01	4.6e+06/ 1.3e+03
has cycle	F	1.3e+08/ 8.7e+07	1.1e+08 /1.5e+08	2.5e+06/ 2.8e+05	5.3e+02/ 1.6e+02	1.8e+05/ 2.2e+04	5.3e+02/ 1.6e+02	5.6e+06/ 2.3e+06
instCountPass	F	5.9e+07/ 9.8e+06	5.6e+07 /5.7e+07	1.2e+06/ 1.1e+04	4.7e+02/ 1.9e+01	2.2e+05/ 1.1e+03	4.7e+02/ 2.0e+01	4.7e+06/ 2.0e+03
latencyModelPass	F	5.8e+07/ 1.5e+07	6.2e+07 /6.9e+07	1.2e+06/ 3.4e+04	6.2e+02/ 6.2e+01	2.0e+05/ 4.0e+03	6.3e+02/ 6.3e+01	4.7e+06/ 5.3e+05
memoryOpStatsPass	F	5.8e+07/ 1.9e+07	8.5e+07 /9.1e+07	6.7e+05/ 3.6e+04	4.7e+02/ 5.4e+01	3.0e+04/ 3.3e+03	4.7e+02/ 5.4e+01	4.6e+06/ 4.4e+05
stripSideEffectsPass	M	1.5e+08 /2.3e+08	1.8e+08 /6.5e+08	8.1e+05 /1.3e+06	1.1e+05 /1.6e+05	7.1e+03 /1.5e+04	4.4e+03 /1.1e+04	4.0e+06/ 1.2e+06
targetReturnPass	M	1.5e+08 /2.2e+08	1.7e+08 /7.1e+08	1.5e+06/ 1.2e+06	1.2e+05 /2.0e+05	7.0e+03 /2.1e+04	6.3e+03 /2.7e+04	4.2e+06/ 1.4e+06
throughputModelPass	F	5.9e+07/ 1.5e+07	6.2e+07 /6.9e+07	1.2e+06/ 2.9e+04	4.6e+02/ 5.1e+01	2.0e+05/ 3.5e+03	4.7e+02/ 5.2e+01	4.7e+06/ 5.3e+05
verifyIR	F	6.1e+07/ 1.7e+07	7.4e+07 /8.6e+07	9.3e+05/ 2.0e+04	5.8e+02/ 1.1e+02	1.4e+05/ 4.2e+03	5.8e+02/ 1.1e+02	4.6e+06/ 4.5e+05
Total		9.3e+08/6.5e+08	9.9e+08/2.1e+09	1.2e+07/3.0e+06	2.4e+05/3.6e+05	1.3e+06/8.0e+04	1.5e+04/3.9e+04	5.1e+07/7.4e+06

Table 8: Per-pass PAPI counters for **Compiler** (Ai/Si). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Ai/Si)	INS (Ai/Si)	L1D (Ai/Si)	L1I (Ai/Si)	L2D (Ai/Si)	L2I (Ai/Si)	LLC (Ai/Si)
blockCountPass	F	4.3e+08/–	2.2e+08/–	2.5e+06/–	1.0e+03/–	1.3e+05/–	1.0e+03/–	1.4e+07/–
branchStatsPass	F	4.9e+08/–	2.5e+08/–	2.7e+06/–	7.2e+02/–	8.7e+04/–	7.1e+02/–	1.7e+07/–
castInstCountPass	F	4.8e+08/–	2.5e+08/–	2.7e+06/–	9.6e+02/–	8.9e+04/–	9.6e+02/–	1.7e+07/–
has cycle	F	4.9e+08/–	2.6e+08/–	2.7e+06/–	4.1e+02/–	8.3e+04/–	4.0e+02/–	1.7e+07/–
instCountPass	F	4.3e+08/–	2.2e+08/–	2.6e+06/–	1.0e+03/–	1.3e+05/–	1.0e+03/–	1.4e+07/–
latencyModelPass	F	4.9e+08/–	2.4e+08/–	2.7e+06/–	7.1e+02/–	8.8e+04/–	7.0e+02/–	1.7e+07/–
memoryOpStatsPass	F	4.8e+08/–	2.6e+08/–	2.7e+06/–	8.6e+02/–	8.8e+04/–	8.6e+02/–	1.7e+07/–
stripSideEffectsPass	M	7.1e+08/–	4.2e+08/–	1.4e+06/–	1.3e+05/–	2.6e+04/–	2.7e+04/–	2.4e+07/–
targetReturnPass	M	6.9e+08/–	4.0e+08/–	1.1e+06/–	1.1e+05/–	3.3e+04/–	3.5e+04/–	2.4e+07/–
throughputModelPass	F	4.8e+08/–	2.4e+08/–	2.7e+06/–	7.0e+02/–	8.7e+04/–	7.0e+02/–	1.7e+07/–
verifyIR	F	3.5e+08/–	1.4e+08/–	2.4e+06/–	8.9e+02/–	1.4e+05/–	8.9e+02/–	1.2e+07/–
Total		5.5e+09/–	2.9e+09/–	2.6e+07/–	2.5e+05/–	9.8e+05/–	7.0e+04/–	1.9e+08/–

Table 9: Per-pass performance for **DBQuery**, ADT has 15 fields, SoA uses 13 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across –iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Si	Am/Sm	Ai/Am	Ai/Sm	Ai/Si
clearQueryFlags	M	14/15	7%	0.0867$\pm$0.00014	0.0976 $\pm$ 0.00127	0.1217 $\pm$ 0.00243	0.3903 $\pm$ 0.00236	0.71 $\times$	1.13$\times$	0.80 $\times$	0.25 $\times$
countJoins	F	3/15	80%	0.0155 $\pm$ 0.00000	0.0324 $\pm$ 0.00001	0.0067$\pm$0.00001	0.1368 $\pm$ 0.00001	2.31$\times$	2.09$\times$	4.84$\times$	0.24 $\times$
scaleCosts	M	15/15	0%	0.0869$\pm$0.00105	0.0962 $\pm$ 0.00012	0.1265 $\pm$ 0.00084	0.3841 $\pm$ 0.00156	0.69 $\times$	1.11$\times$	0.76 $\times$	0.25 $\times$
sumCost	F	6/15	60%	0.0158 $\pm$ 0.00007	0.0301 $\pm$ 0.00100	0.0104$\pm$0.00006	0.1562 $\pm$ 0.00118	1.52$\times$	1.91$\times$	2.89$\times$	0.19 $\times$
sumMemory	F	6/15	60%	0.0156 $\pm$ 0.00001	0.0302 $\pm$ 0.00001	0.0107$\pm$0.00002	0.1540 $\pm$ 0.00182	1.46$\times$	1.94$\times$	2.82$\times$	0.20 $\times$
sumRows	F	6/15	60%	0.0158 $\pm$ 0.00001	0.0302 $\pm$ 0.00081	0.0106$\pm$0.00004	0.1570 $\pm$ 0.00018	1.49$\times$	1.91$\times$	2.84$\times$	0.19 $\times$
Total				0.2362	0.3166	0.2866	1.378	0.82 $\times$	1.34$\times$	1.10$\times$	0.23 $\times$
Geomean								1.24$\times$	1.62$\times$	2.02$\times$	0.22 $\times$

Table 10: Per-pass PAPI counters for **DBQuery** (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
clearQueryFlags	M	1.9e+08 /3.6e+08	3.2e+08 /1.3e+09	6.4e+05 /2.0e+06	1.5e+05 /3.0e+05	8.1e+03 /2.9e+04	5.6e+03 /4.8e+04	3.6e+06/ 1.4e+06
countJoins	F	7.8e+07/ 3.4e+07	1.3e+08/ 1.2e+08	2.4e+06/ 4.2e+04	5.5e+01 /6.9e+01	2.1e+04/ 1.6e+03	6.4e+01 /6.8e+01	4.5e+06/ 4.0e+03
scaleCosts	M	1.9e+08 /3.8e+08	3.8e+08 /1.4e+09	1.2e+06 /2.1e+06	1.1e+05 /2.5e+05	1.1e+04 /2.9e+04	5.1e+03 /4.7e+04	3.6e+06/ 1.4e+06
sumCost	F	8.0e+07/ 5.2e+07	1.5e+08 /2.1e+08	2.1e+06/ 1.2e+05	1.4e+02/ 8.9e+01	2.2e+04/ 4.5e+03	1.4e+02/ 9.2e+01	4.7e+06/ 4.8e+05
sumMemory	F	7.9e+07/ 5.4e+07	1.5e+08 /2.1e+08	2.1e+06/ 2.0e+05	6.4e+01 /6.9e+01	2.1e+04/ 4.5e+03	6.6e+01 /6.9e+01	4.7e+06/ 4.7e+05
sumRows	F	8.0e+07/ 5.4e+07	1.5e+08 /2.1e+08	2.1e+06/ 1.6e+05	8.3e+01/ 7.3e+01	1.4e+04/ 4.5e+03	9.1e+01/ 7.6e+01	4.7e+06/ 5.0e+05
Total		6.9e+08/9.3e+08	1.3e+09/3.5e+09	1.0e+07/4.7e+06	2.6e+05/5.6e+05	9.7e+04/7.3e+04	1.1e+04/9.5e+04	2.6e+07/4.2e+06

Table 11: Per-pass PAPI counters for **DBQuery** (Ai/Si). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Ai/Si)	INS (Ai/Si)	L1D (Ai/Si)	L1I (Ai/Si)	L2D (Ai/Si)	L2I (Ai/Si)	LLC (Ai/Si)
clearQueryFlags	M	2.4e+08 /1.7e+09	5.6e+08 /3.7e+09	5.1e+05 /1.0e+07	1.5e+05 /6.1e+05	4.7e+03 /5.0e+04	5.7e+03 /9.2e+04	3.3e+06/ 1.4e+06
countJoins	F	1.6e+08 /7.0e+08	3.5e+08 /1.2e+09	2.9e+06/ 1.7e+05	1.2e+02 /2.4e+03	1.2e+04 /1.2e+04	1.3e+02 /1.6e+02	3.1e+06/ 2.0e+04
scaleCosts	M	2.4e+08 /1.7e+09	6.0e+08 /3.7e+09	6.2e+05 /1.2e+07	1.2e+05 /9.0e+05	4.6e+03 /6.3e+04	6.1e+03 /8.1e+04	3.2e+06/ 1.4e+06
sumCost	F	1.5e+08 /7.9e+08	3.8e+08 /1.2e+09	9.4e+05 /1.2e+06	8.6e+01 /2.5e+03	1.7e+03 /2.5e+04	8.7e+01 /5.7e+02	3.4e+06/ 4.0e+05
sumMemory	F	1.5e+08 /7.8e+08	3.8e+08 /1.2e+09	1.1e+06 /1.3e+06	6.0e+01 /2.2e+03	1.6e+03 /2.7e+04	6.1e+01 /7.4e+02	3.4e+06/ 3.9e+05
sumRows	F	1.5e+08 /8.0e+08	3.8e+08 /1.2e+09	1.1e+06 /1.3e+06	1.9e+02 /2.3e+03	1.7e+03 /2.7e+04	1.9e+02 /6.1e+02	3.3e+06/ 4.0e+05
Total		1.1e+09/6.5e+09	2.6e+09/1.2e+10	7.1e+06/2.6e+07	2.6e+05/1.5e+06	2.6e+04/2.0e+05	1.2e+04/1.7e+05	2.0e+07/4.0e+06

Table 12: Per-pass performance for **DecisionTree**, ADT has 7 fields, SoA uses 6 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across –iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Si	Am/Sm	Ai/Am	Ai/Sm	Ai/Si
classify Batch	F	5/7	29%	3.181 $\pm$ 0.0034	4.260 $\pm$ 0.0048	2.627$\pm$0.0030	24.35 $\pm$ 0.010	1.21$\times$	1.34$\times$	1.62$\times$	0.17 $\times$
classify Depth	F	4/7	43%	0.0458 $\pm$ 0.00002	0.0609 $\pm$ 0.00002	0.0373$\pm$0.00015	0.3472 $\pm$ 0.00166	1.23$\times$	1.33$\times$	1.63$\times$	0.18 $\times$
classify tree	F	5/7	29%	0.0457 $\pm$ 0.00003	0.0609 $\pm$ 0.00019	0.0372$\pm$0.00005	0.3470 $\pm$ 0.00019	1.23$\times$	1.33$\times$	1.64$\times$	0.18 $\times$
countFeatureUses	F	3/7	57%	0.0502 $\pm$ 0.00002	0.1429 $\pm$ 0.00132	0.0318$\pm$0.00003	0.3927 $\pm$ 0.00120	1.58$\times$	2.84$\times$	4.50$\times$	0.36 $\times$
countLeaves	F	2/7	71%	0.0498 $\pm$ 0.00001	0.1459 $\pm$ 0.00102	0.0309$\pm$0.00001	0.3736 $\pm$ 0.00101	1.61$\times$	2.93$\times$	4.72$\times$	0.39 $\times$
countNodes	F	2/7	71%	0.0494 $\pm$ 0.00002	0.1428 $\pm$ 0.00016	0.0313$\pm$0.00100	0.3751 $\pm$ 0.00118	1.58$\times$	2.89$\times$	4.56$\times$	0.38 $\times$
countSmallLeaves	F	3/7	57%	0.0502 $\pm$ 0.00002	0.1431 $\pm$ 0.00155	0.0323$\pm$0.00004	0.3225 $\pm$ 0.00140	1.55$\times$	2.85$\times$	4.43$\times$	0.44 $\times$
inferenceCost	F	2/7	71%	0.0499 $\pm$ 0.00158	0.1458 $\pm$ 0.00173	0.0308$\pm$0.00001	0.3762 $\pm$ 0.00225	1.62$\times$	2.92$\times$	4.73$\times$	0.39 $\times$
sumImpurity	F	3/7	57%	0.0498 $\pm$ 0.00002	0.1407 $\pm$ 0.00130	0.0312$\pm$0.00003	0.3843 $\pm$ 0.00173	1.60$\times$	2.82$\times$	4.51$\times$	0.37 $\times$
sumPathLengths	F	3/7	57%	0.0510 $\pm$ 0.00001	0.1455 $\pm$ 0.00003	0.0333$\pm$0.00002	0.3773 $\pm$ 0.00271	1.53$\times$	2.85$\times$	4.37$\times$	0.39 $\times$
sumSamples	F	3/7	57%	0.0491 $\pm$ 0.00001	0.1402 $\pm$ 0.00015	0.0319$\pm$0.00119	0.3787 $\pm$ 0.00202	1.54$\times$	2.86$\times$	4.40$\times$	0.37 $\times$
treeDepth	F	2/7	71%	0.0502 $\pm$ 0.00018	0.1456 $\pm$ 0.00151	0.0335$\pm$0.00002	0.3759 $\pm$ 0.00134	1.50$\times$	2.90$\times$	4.34$\times$	0.39 $\times$
Total				3.722	5.674	2.989	28.40	1.25$\times$	1.52$\times$	1.90$\times$	0.20 $\times$
Geomean								1.47$\times$	2.37$\times$	3.49$\times$	0.32 $\times$

Table 13: Per-pass PAPI counters for **DecisionTree** (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
classify Batch	F	1.6e+10/ 1.3e+10	3.6e+10 /5.5e+10	5.1e+08/ 2.8e+07	1.4e+05/ 5.0e+04	1.4e+06 /2.0e+06	1.4e+05/ 4.8e+04	9.2e+08/ 1.8e+07
classify Depth	F	2.3e+08/ 1.9e+08	5.1e+08 /7.8e+08	7.4e+06/ 4.4e+05	2.1e+03/ 5.8e+02	2.0e+04 /2.8e+04	2.1e+03/ 5.5e+02	1.3e+07/ 2.8e+05
classify tree	F	2.3e+08/ 1.9e+08	5.1e+08 /7.8e+08	7.4e+06/ 4.3e+05	2.0e+03/ 5.8e+02	1.9e+04 /2.8e+04	2.0e+03/ 5.6e+02	1.3e+07/ 2.8e+05
countFeatureUses	F	2.6e+08/ 1.6e+08	7.7e+08 /8.4e+08	6.1e+06/ 4.0e+05	2.6e+03/ 4.5e+02	3.1e+04/ 1.2e+04	2.6e+03/ 4.6e+02	1.5e+07/ 2.1e+06
countLeaves	F	2.5e+08/ 1.6e+08	5.7e+08/ 5.5e+08	6.2e+06/ 1.2e+05	3.1e+03/ 1.2e+03	3.6e+04/ 8.6e+03	3.1e+03/ 1.1e+03	1.4e+07/ 3.5e+05
countNodes	F	2.5e+08/ 1.6e+08	5.7e+08/ 5.5e+08	6.4e+06/ 1.2e+05	2.8e+03/ 1.2e+03	3.7e+04/ 8.6e+03	2.8e+03/ 1.1e+03	1.4e+07/ 3.6e+05
countSmallLeaves	F	2.5e+08/ 1.6e+08	6.6e+08 /7.8e+08	5.3e+06/ 6.9e+04	2.3e+03/ 4.7e+02	3.2e+04/ 1.3e+04	2.3e+03/ 4.7e+02	1.5e+07/ 2.0e+06
inferenceCost	F	2.5e+08/ 1.6e+08	6.9e+08/ 6.9e+08	9.0e+06/ 1.5e+05	1.9e+03/ 7.6e+02	2.0e+04/ 8.1e+03	1.9e+03/ 7.3e+02	1.4e+07/ 3.4e+05
sumImpurity	F	2.5e+08/ 1.6e+08	7.0e+08 /7.8e+08	6.6e+06/ 3.2e+05	2.5e+03/ 8.3e+02	3.2e+04/ 1.2e+04	2.5e+03/ 8.2e+02	1.5e+07/ 2.0e+06
sumPathLengths	F	2.6e+08/ 1.7e+08	7.4e+08 /8.7e+08	7.5e+06/ 7.3e+05	3.1e+03/ 6.3e+02	3.5e+04/ 1.3e+04	3.1e+03/ 6.3e+02	1.5e+07/ 2.0e+06
sumSamples	F	2.5e+08/ 1.6e+08	5.9e+08 /7.1e+08	6.0e+06/ 2.0e+05	2.3e+03/ 1.5e+03	3.4e+04/ 1.5e+04	2.3e+03/ 1.5e+03	1.5e+07/ 2.0e+06
treeDepth	F	2.5e+08/ 1.7e+08	6.9e+08/ 6.9e+08	8.7e+06/ 1.4e+05	1.5e+03/ 5.9e+02	1.7e+04/ 7.9e+03	1.5e+03/ 5.9e+02	1.4e+07/ 3.3e+05
Total		1.9e+10/1.5e+10	4.3e+10/6.3e+10	5.9e+08/3.1e+07	1.6e+05/5.9e+04	1.7e+06/2.1e+06	1.6e+05/5.6e+04	1.1e+09/3.0e+07

Table 14: Per-pass PAPI counters for **DecisionTree** (Ai/Si). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Ai/Si)	INS (Ai/Si)	L1D (Ai/Si)	L1I (Ai/Si)	L2D (Ai/Si)	L2I (Ai/Si)	LLC (Ai/Si)
classify Batch	F	2.2e+10 /1.2e+11	6.6e+10 /2.4e+11	4.7e+07 /4.8e+07	3.4e+04/ 3.2e+04	4.3e+05 /2.2e+06	3.4e+04/ 2.6e+04	8.6e+08/ 1.8e+07
classify Depth	F	3.1e+08 /1.8e+09	9.4e+08 /3.4e+09	8.0e+05/ 6.9e+05	4.2e+02/ 3.2e+02	4.7e+03 /3.2e+04	4.3e+02/ 2.9e+02	1.2e+07/ 2.9e+05
classify tree	F	3.1e+08 /1.8e+09	9.4e+08 /3.4e+09	7.9e+05/ 6.9e+05	3.6e+02/ 3.6e+02	4.5e+03 /3.2e+04	3.6e+02/ 3.1e+02	1.2e+07/ 3.0e+05
countFeatureUses	F	7.2e+08 /2.0e+09	1.9e+09 /4.0e+09	2.1e+06 /3.7e+06	3.4e+02 /1.1e+03	5.4e+03 /7.4e+04	3.4e+02 /4.8e+02	9.2e+06/ 1.1e+06
countLeaves	F	7.4e+08 /1.9e+09	1.7e+09 /3.8e+09	6.1e+06/ 7.3e+05	4.2e+02 /2.2e+03	3.1e+04 /3.4e+04	4.2e+02/ 2.4e+02	9.1e+06/ 3.2e+05
countNodes	F	7.2e+08 /1.9e+09	1.7e+09 /3.8e+09	3.2e+06/ 7.4e+05	1.8e+03/ 1.5e+03	2.2e+04 /3.4e+04	1.8e+03/ 2.2e+02	9.2e+06/ 3.2e+05
countSmallLeaves	F	7.3e+08 /1.6e+09	1.8e+09 /3.9e+09	1.2e+06 /3.6e+06	3.7e+02 /1.9e+03	9.6e+03 /7.7e+04	3.8e+02 /1.2e+03	9.1e+06/ 1.2e+06
inferenceCost	F	7.4e+08 /1.9e+09	1.8e+09 /3.8e+09	5.2e+06/ 7.2e+05	8.5e+02 /2.9e+03	2.7e+04 /3.4e+04	8.6e+02/ 2.9e+02	9.1e+06/ 3.0e+05
sumImpurity	F	7.1e+08 /2.0e+09	1.8e+09 /3.8e+09	1.1e+06 /3.6e+06	5.7e+02/ 3.8e+02	5.6e+03 /7.2e+04	5.7e+02/ 3.5e+02	9.3e+06/ 1.1e+06
sumPathLengths	F	7.4e+08 /1.9e+09	1.9e+09 /3.9e+09	1.5e+06 /3.7e+06	7.2e+02 /9.2e+02	5.6e+03 /7.8e+04	7.2e+02/ 6.6e+02	9.1e+06/ 1.1e+06
sumSamples	F	7.1e+08 /1.9e+09	1.7e+09 /3.8e+09	8.9e+05 /3.7e+06	6.2e+02 /1.1e+03	9.1e+03 /7.8e+04	6.2e+02/ 3.8e+02	9.2e+06/ 1.1e+06
treeDepth	F	7.4e+08 /1.9e+09	1.8e+09 /3.8e+09	5.2e+06/ 7.2e+05	3.6e+03/ 1.9e+03	3.0e+04 /3.4e+04	3.6e+03/ 3.1e+02	9.0e+06/ 3.1e+05
Total		2.9e+10/1.4e+11	8.4e+10/2.8e+11	7.5e+07/7.1e+07	4.4e+04/4.7e+04	5.8e+05/2.8e+06	4.4e+04/3.1e+04	9.7e+08/2.6e+07

Table 15: Per-pass performance for **DomTree**, ADT has 15 fields, SoA uses 14 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across –iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Si	Am/Sm	Ai/Am	Ai/Sm	Ai/Si
SumArea	F	6/15	60%	0.0343 $\pm$ 0.00002	0.0700 $\pm$ 0.00002	0.0185$\pm$0.00006	0.1957 $\pm$ 0.00115	1.85$\times$	2.04$\times$	3.77$\times$	0.36 $\times$
computeWidths	M	14/15	7%	0.0301$\pm$0.00003	0.0314 $\pm$ 0.00004	0.0504 $\pm$ 0.00006	0.0989 $\pm$ 0.00007	0.60 $\times$	1.04 $\times$	0.62 $\times$	0.32 $\times$
count styled	F	3/15	80%	0.0334 $\pm$ 0.00001	0.0690 $\pm$ 0.00013	0.0116$\pm$0.00002	0.1853 $\pm$ 0.00130	2.89$\times$	2.06$\times$	5.96$\times$	0.37 $\times$
find max Bottom	F	5/15	67%	0.0332 $\pm$ 0.00001	0.0684 $\pm$ 0.00002	0.0201$\pm$0.00001	0.1971 $\pm$ 0.00119	1.65$\times$	2.06$\times$	3.41$\times$	0.35 $\times$
scaleLayout	M	15/15	0%	0.0284$\pm$0.00005	0.0309 $\pm$ 0.00106	0.0391 $\pm$ 0.00004	0.0912 $\pm$ 0.00163	0.73 $\times$	1.09 $\times$	0.79 $\times$	0.34 $\times$
sumTextWidth	F	3/15	80%	0.0332 $\pm$ 0.00000	0.0680 $\pm$ 0.00002	0.0116$\pm$0.00009	0.2552 $\pm$ 0.00292	2.87$\times$	2.05$\times$	5.88$\times$	0.27 $\times$
Total				0.1926	0.3376	0.1512	1.023	1.27$\times$	1.75$\times$	2.23$\times$	0.33 $\times$
Geomean								1.49$\times$	1.65$\times$	2.46$\times$	0.33 $\times$

Table 16: Per-pass PAPI counters for **DomTree** (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
SumArea	F	1.7e+08/ 9.4e+07	2.7e+08 /4.2e+08	2.4e+06/ 5.6e+05	2.0e+03/ 5.8e+01	1.8e+05/ 1.2e+04	2.0e+03/ 5.9e+01	1.4e+07/ 1.9e+06
computeWidths	M	6.6e+07 /1.7e+08	1.4e+08 /5.8e+08	1.2e+06 /1.4e+06	4.2e+04 /6.6e+04	3.9e+03 /9.5e+03	1.6e+03 /1.7e+04	1.2e+06/ 4.2e+05
count styled	F	1.7e+08/ 5.8e+07	2.5e+08 /2.6e+08	2.0e+06/ 2.6e+04	2.1e+03/ 5.8e+01	2.6e+05/ 5.0e+03	2.0e+03/ 6.1e+01	1.4e+07/ 6.5e+05
find max Bottom	F	1.7e+08/ 1.0e+08	2.6e+08 /4.7e+08	2.2e+06/ 8.2e+05	4.1e+02/ 1.2e+02	1.7e+05/ 9.2e+03	4.3e+02/ 1.2e+02	1.4e+07/ 1.2e+06
scaleLayout	M	5.9e+07 /1.1e+08	8.8e+07 /3.6e+08	1.0e+06 /1.3e+06	7.6e+04/ 5.4e+04	1.5e+04/ 9.2e+03	4.1e+03 /1.6e+04	1.3e+06/ 4.5e+05
sumTextWidth	F	1.7e+08/ 5.9e+07	2.0e+08 /2.4e+08	2.4e+06/ 1.9e+04	1.8e+03/ 9.5e+01	4.3e+05/ 4.3e+03	1.8e+03/ 9.9e+01	1.4e+07/ 6.5e+05
Total		8.1e+08/5.9e+08	1.2e+09/2.3e+09	1.1e+07/4.1e+06	1.2e+05/1.2e+05	1.1e+06/4.9e+04	1.2e+04/3.3e+04	5.7e+07/5.2e+06

Table 17: Per-pass PAPI counters for **DomTree** (Ai/Si). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Ai/Si)	INS (Ai/Si)	L1D (Ai/Si)	L1I (Ai/Si)	L2D (Ai/Si)	L2I (Ai/Si)	LLC (Ai/Si)
SumArea	F	3.6e+08 /9.9e+08	6.4e+08 /2.1e+09	8.5e+06/ 3.3e+06	3.5e+02 /1.9e+03	1.1e+04 /5.1e+04	3.5e+02 /1.8e+03	1.0e+07/ 1.3e+06
computeWidths	M	7.3e+07 /4.1e+08	1.8e+08 /1.0e+09	1.3e+06 /3.5e+06	5.6e+04 /8.8e+04	4.2e+03 /1.6e+04	1.7e+03 /2.3e+04	1.1e+06/ 4.9e+05
count styled	F	3.5e+08 /9.4e+08	6.3e+08 /2.1e+09	1.0e+07/ 1.3e+06	2.2e+02 /1.5e+03	1.4e+04 /3.9e+04	2.2e+02 /3.1e+02	1.0e+07/ 4.2e+05

Table 18: Per-pass performance for **KDTree**, ADT has 19 fields, SoA uses 12 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across $-iterate$ runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Si	Am/Sm	Ai/Am	Ai/Sm	Ai/Si
Find nearest Neighbour	F	11/19	42%	0.3661 $\pm$ 0.00105	0.4402 $\pm$ 0.00111	0.2895$\pm$0.00017	2.626 $\pm$ 0.0046	1.26$\times$	1.20$\times$	1.52$\times$	0.17 $\times$
Total				0.3661	0.4402	0.2895	2.626	1.26$\times$	1.20$\times$	1.52$\times$	0.17 $\times$
Geomean								1.26$\times$	1.20$\times$	1.52$\times$	0.17 $\times$

Table 19: Per-pass PAPI counters for **KDTree** (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
Find nearest Neighbour	F	1.9e+09/ 1.5e+09	6.9e+09 /8.0e+09	2.2e+07/ 3.5e+06	1.2e+03/ 4.7e+02	4.3e+04 /1.3e+05	1.2e+03/ 4.7e+02	8.1e+07/ 2.3e+07
Total		1.9e+09/1.5e+09	6.9e+09/8.0e+09	2.2e+07/3.5e+06	1.2e+03/4.7e+02	4.3e+04/1.3e+05	1.2e+03/4.7e+02	8.1e+07/2.3e+07

Table 20: Per-pass PAPI counters for **KDTree** (Ai/Si). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Ai/Si)	INS (Ai/Si)	L1D (Ai/Si)	L1I (Ai/Si)	L2D (Ai/Si)	L2I (Ai/Si)	LLC (Ai/Si)
Find nearest Neighbour	F	2.2e+09 /1.3e+10	8.9e+09 /2.2e+10	1.5e+07 /2.9e+07	2.3e+03 /3.0e+04	3.0e+04 /6.0e+05	2.3e+03 /2.6e+04	8.1e+07/ 1.6e+07
Total		2.2e+09/1.3e+10	8.9e+09/2.2e+10	1.5e+07/2.9e+07	2.3e+03/3.0e+04	3.0e+04/6.0e+05	2.3e+03/2.6e+04	8.1e+07/1.6e+07

Table 21: Per-pass performance for **LinearListReduction**, ADT has 11 fields, SoA uses 11 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across $-iterate$ runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Si	Am/Sm	Ai/Am	Ai/Sm	Ai/Si
reduction	F	2/11	82%	0.0274 $\pm$ 0.00002	0.1652 $\pm$ 0.00082	0.0049$\pm$0.00001	<i>OOM</i>	5.59$\times$	6.03$\times$	33.68$\times$	–
Total				0.0274	0.1652	0.0049	–	5.59$\times$	6.03$\times$	33.68$\times$	–
Geomean								5.59$\times$	6.03$\times$	33.68$\times$	–

Table 22: Per-pass PAPI counters for **LinearListReduction** (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
reduction	F	1.4e+08/ 2.5e+07	9.0e+07 /1.0e+08	4.7e+06/ 5.8e+04	7.3e+02/ 4.6e+01	1.9e+06/ 7.2e+03	7.3e+02/ 4.7e+01	1.2e+07/ 1.1e+06
Total		1.4e+08/2.5e+07	9.0e+07/1.0e+08	4.7e+06/5.8e+04	7.3e+02/4.6e+01	1.9e+06/7.2e+03	7.3e+02/4.7e+01	1.2e+07/1.1e+06

Table 23: Per-pass PAPI counters for **LinearListReduction** (Ai/Si). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Ai/Si)	INS (Ai/Si)	L1D (Ai/Si)	L1I (Ai/Si)	L2D (Ai/Si)	L2I (Ai/Si)	LLC (Ai/Si)
reduction	F	8.4e+08/–	3.7e+08/–	5.0e+06/–	2.5e+03/–	3.3e+05/–	2.5e+03/–	3.0e+07/–
Total		8.4e+08/–	3.7e+08/–	5.0e+06/–	2.5e+03/–	3.3e+05/–	2.5e+03/–	3.0e+07/–

Table 24: Per-pass performance for **List**, ADT has 2 fields, SoA uses 2 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across $-iterate$ runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Si	Am/Sm	Ai/Am	Ai/Sm	Ai/Si
add1 List	M	2/2	0%	0.2532$\pm$0.00066	<i>OOM</i>	0.3113 $\pm$ 0.00290	<i>OOM</i>	0.81 $\times$	–	–	–
length List	M	1/2	50%	0.0411 $\pm$ 0.00001	<i>OOM</i>	0.0216$\pm$0.00005	<i>OOM</i>	1.91$\times$	–	–	–
sumList	F	2/2	0%	0.0439$\pm$0.00092	<i>OOM</i>	0.0455 $\pm$ 0.00044	<i>OOM</i>	0.96 $\times$	–	–	–
sumList tail recursive	F	2/2	0%	0.0438$\pm$0.00014	<i>OOM</i>	0.0451 $\pm$ 0.00006	<i>OOM</i>	0.97 $\times$	–	–	–
Total				0.3820	–	0.4234	–	0.90 $\times$	–	–	–
Geomean								1.10 $\times$	–	–	–

Table 25: Per-pass PAPI counters for **List** (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
add1 List	M	6.0e+08 /8.4e+08	1.9e+09 /3.1e+09	7.7e+06/ 1.1e+06	3.0e+05 /6.1e+05	1.1e+04 /5.5e+04	1.1e+04 /1.9e+04	9.1e+06/ 8.1e+06
length List	M	2.1e+08/ 1.1e+08	8.0e+08/ 8.0e+08	1.2e+05 /2.8e+05	2.1e+02/ 7.6e+01	3.2e+03 /1.8e+04	2.1e+02/ 7.8e+01	1.3e+07/ 1.2e+06
sumList	F	2.2e+08 /2.3e+08	9.0e+08 /1.1e+09	9.2e+05/ 6.2e+05	2.0e+02/ 1.2e+02	3.9e+03 /5.8e+04	2.0e+02/ 1.2e+02	1.3e+07/ 1.2e+07
sumList tail recursive	F	2.2e+08 /2.3e+08	9.0e+08 /1.1e+09	9.2e+05/ 7.0e+05	1.8e+02/ 8.9e+01	3.9e+03 /5.2e+04	1.8e+02/ 9.0e+01	1.3e+07/ 1.2e+07
Total		1.2e+09/1.4e+09	4.5e+09/6.1e+09	9.7e+06/2.7e+06	3.0e+05/6.1e+05	2.2e+04/1.8e+05	1.1e+04/1.9e+04	4.8e+07/3.4e+07

Table 26: Per-pass PAPI counters for **List** (Ai/Si). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Ai/Si)	INS (Ai/Si)	L1D (Ai/Si)	L1I (Ai/Si)	L2D (Ai/Si)	L2I (Ai/Si)	LLC (Ai/Si)
Total		–/–	–/–	–/–	–/–	–/–	–/–	–/–

Table 27: Per-pass performance for **MonoTree**, ADT has 3 fields, SoA uses 2 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across $-iterate$ runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Si	Am/Sm	Ai/Am	Ai/Sm	Ai/Si
add1Tree	M	3/3	0%	0.0462 $\pm$ 0.00114	0.0641 $\pm$ 0.00028	0.0421$\pm$0.00119	0.1298 $\pm$ 0.00114	1.10 $\times$	1.39$\times$	1.52$\times$	0.49 $\times$
sumTree	F	3/3	0%	0.0116 $\pm$ 0.00000	0.0424 $\pm$ 0.00128	0.0113$\pm$0.00001	0.0818 $\pm$ 0.00002	1.02 $\times$	3.66$\times$	3.74$\times$	0.52 $\times$
sumTree TailRec	F	3/3	0%	0.0120 $\pm$ 0.00001	0.0425 $\pm$ 0.00002	0.0111$\pm$0.00002	0.0824 $\pm$ 0.00018	1.08 $\times$	3.53$\times$	3.81$\times$	0.52 $\times$
Total				0.0698	0.1490	0.0645	0.2941	1.08 $\times$	2.14$\times$	2.31$\times$	0.51 $\times$
Geomean								1.07 $\times$	2.62$\times$	2.79$\times$	0.51 $\times$

Table 28: Per-pass PAPI counters for `MonoTree` (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
add1Tree	M	1.7e+08/ 1.4e+08	5.0e+08 /6.4e+08	1.6e+05 /4.4e+05	5.3e+04 /8.0e+04	2.4e+03 /6.1e+03	4.1e+03 /5.0e+03	5.0e+05/ 4.5e+05
sumTree	F	5.9e+07/ 5.8e+07	2.1e+08 /2.5e+08	4.0e+03 /5.2e+04	2.0e+02 /5.9e+02	4.5e+02 /4.3e+03	2.1e+02 /5.9e+02	7.1e+05/ 6.2e+05
sumTree TailRec	F	6.1e+07/ 5.7e+07	1.9e+08 /2.2e+08	4.0e+03 /3.7e+04	2.1e+02 /5.2e+02	5.0e+02 /4.1e+03	2.1e+02 /5.2e+02	6.9e+05/ 6.1e+05
Total		2.9e+08/2.6e+08	9.1e+08/1.1e+09	1.7e+05/5.3e+05	5.3e+04/8.1e+04	3.3e+03/1.5e+04	4.5e+03/6.1e+03	1.9e+06/1.7e+06

Table 29: Per-pass PAPI counters for `MonoTree` (Ai/Si). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Ai/Si)	INS (Ai/Si)	L1D (Ai/Si)	L1I (Ai/Si)	L2D (Ai/Si)	L2I (Ai/Si)	LLC (Ai/Si)
add1Tree	M	2.6e+08 /5.9e+08	9.1e+08 /1.4e+09	2.1e+05 /1.1e+06	3.6e+04 /6.6e+04	1.6e+03 /2.2e+04	2.3e+03/ 2.0e+03	4.7e+05/ 3.8e+05
sumTree	F	2.2e+08 /4.2e+08	6.0e+08 /7.6e+08	8.4e+04 /1.2e+06	2.5e+02 /3.7e+02	6.9e+02 /2.0e+04	2.5e+02 /2.6e+02	3.4e+05 /3.6e+05
sumTree TailRec	F	2.2e+08 /4.2e+08	6.0e+08 /7.6e+08	8.6e+04 /1.2e+06	1.6e+02 /2.7e+02	7.4e+02 /2.0e+04	1.6e+02 /2.6e+02	3.3e+05 /3.6e+05
Total		6.9e+08/1.4e+09	2.1e+09/2.9e+09	3.8e+05/3.4e+06	3.6e+04/6.7e+04	3.0e+03/6.2e+04	2.7e+03/2.5e+03	1.1e+06/1.1e+06

Table 30: Per-pass performance for `ObjectGraph`, ADT has 5 fields, SoA uses 4 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across -iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Si	Am/Sm	Ai/Am	Ai/Sm	Ai/Si
clearMarks	M	4/5	20%	0.0094$\pm$0.00001	0.0114 $\pm$ 0.00001	0.0101 $\pm$ 0.00001	0.0264 $\pm$ 0.00002	0.93 $\times$	1.22$\times$	1.13$\times$	0.43 $\times$
countLargeItems	F	3/5	40%	0.0017 $\pm$ 0.00000	0.0055 $\pm$ 0.00000	0.0013$\pm$0.00000	0.0064 $\pm$ 0.00000	1.26$\times$	3.31$\times$	4.16$\times$	0.86 $\times$
countMarked	F	3/5	40%	0.0017 $\pm$ 0.00000	0.0055 $\pm$ 0.00000	0.0013$\pm$0.00000	0.0061 $\pm$ 0.00000	1.31$\times$	3.30$\times$	4.33$\times$	0.90 $\times$
inflateSizes	M	5/5	0%	0.0090$\pm$0.00001	0.0114 $\pm$ 0.00001	0.0099 $\pm$ 0.00001	0.0270 $\pm$ 0.00001	0.91 $\times$	1.26$\times$	1.15$\times$	0.42 $\times$
sumObjIds	F	3/5	40%	0.0016 $\pm$ 0.00001	0.0054 $\pm$ 0.00000	0.0012$\pm$0.00000	0.0062 $\pm$ 0.00000	1.31$\times$	3.30$\times$	4.33$\times$	0.86 $\times$
totalHeapSize	F	3/5	40%	0.0016 $\pm$ 0.00024	0.0054 $\pm$ 0.00048	0.0013$\pm$0.00023	0.0062 $\pm$ 0.00039	1.28$\times$	3.26$\times$	4.17$\times$	0.87 $\times$
Total				0.0250	0.0445	0.0251	0.0784	1.00 $\times$	1.78$\times$	1.77$\times$	0.57 $\times$
Geomean								1.15$\times$	2.38$\times$	2.74$\times$	0.69 $\times$

Table 31: Per-pass PAPI counters for `ObjectGraph` (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
clearMarks	M	2.6e+07 /2.9e+07	6.1e+07 /1.1e+08	9.4e+04/ 4.4e+04	1.2e+04 /1.4e+04	7.1e+02 /1.1e+03	1.2e+03/ 6.3e+02	1.1e+05/ 5.1e+04
countLargeItems	F	8.5e+06/ 6.7e+06	3.5e+07 /3.9e+07	4.1e+04/ 8.3e+03	8.5e+01/ 2.6e+01	2.8e+02 /5.2e+02	8.4e+01/ 2.8e+01	7.6e+04/ 1.5e+03
countMarked	F	8.5e+06/ 6.5e+06	3.5e+07 /3.9e+07	4.2e+04/ 6.5e+03	8.1e+01/ 2.9e+01	2.6e+02 /5.6e+02	8.3e+01/ 3.0e+01	7.5e+04/ 5.3e+02
inflateSizes	M	2.4e+07 /2.8e+07	6.6e+07 /1.2e+08	7.4e+04/ 5.7e+04	1.5e+04/ 1.4e+04	8.0e+02 /1.3e+03	1.4e+03/ 7.1e+02	1.2e+05/ 6.2e+04
sumObjIds	F	8.3e+06/ 6.3e+06	3.1e+07 /3.6e+07	6.0e+04/ 4.6e+03	1.0e+02/ 2.7e+01	3.3e+02 /5.5e+02	1.0e+02/ 2.9e+01	7.7e+04/ 4.8e+02
totalHeapSize	F	8.2e+06/ 6.4e+06	3.1e+07 /3.6e+07	5.9e+04/ 5.8e+03	8.3e+01/ 3.7e+01	3.2e+02 /5.2e+02	8.4e+01/ 3.8e+01	7.7e+04/ 2.4e+03
Total		8.3e+07/8.4e+07	2.6e+08/3.8e+08	3.7e+05/1.3e+05	2.7e+04/2.8e+04	2.7e+03/4.6e+03	2.9e+03/1.5e+03	5.3e+05/1.2e+05

Table 32: Per-pass PAPI counters for `ObjectGraph` (Ai/Si). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Ai/Si)	INS (Ai/Si)	L1D (Ai/Si)	L1I (Ai/Si)	L2D (Ai/Si)	L2I (Ai/Si)	LLC (Ai/Si)
clearMarks	M	3.6e+07 /1.1e+08	1.1e+08 /2.9e+08	3.4e+04 /2.7e+05	9.2e+03 /1.8e+04	5.9e+02 /3.2e+03	5.2e+02 /6.0e+02	1.0e+05/ 6.1e+04
countLargeItems	F	2.8e+07 /3.3e+07	8.2e+07 /1.4e+08	1.8e+04 /1.3e+05	5.1e+01/ 3.5e+01	2.5e+02 /2.0e+03	5.6e+01/ 3.6e+01	7.8e+04/ 1.8e+03
countMarked	F	2.8e+07 /3.1e+07	8.2e+07 /1.4e+08	1.7e+04 /1.2e+05	5.3e+01 /8.5e+01	2.1e+02 /1.7e+03	5.3e+01 /8.5e+01	7.6e+04/ 1.7e+03
inflateSizes	M	3.6e+07 /1.2e+08	1.2e+08 /3.0e+08	1.8e+04 /3.6e+05	1.1e+04 /2.0e+04	4.3e+02 /3.3e+03	4.3e+02 /1.1e+03	1.1e+05/ 8.0e+04
sumObjIds	F	2.7e+07 /3.2e+07	8.0e+07 /1.4e+08	1.7e+04 /1.3e+05	5.3e+01 /9.8e+01	2.0e+02 /2.4e+03	5.4e+01 /9.7e+01	7.7e+04/ 9.2e+02
totalHeapSize	F	2.7e+07 /3.1e+07	8.0e+07 /1.4e+08	1.7e+04 /1.2e+05	5.0e+01 /5.2e+01	2.1e+02 /2.0e+03	5.2e+01/5.2e+01	7.6e+04/ 2.3e+03
Total		1.8e+08/3.5e+08	5.6e+08/1.1e+09	1.2e+05/1.1e+06	2.0e+04/3.8e+04	1.9e+03/1.4e+04	1.2e+03/1.9e+03	5.2e+05/1.5e+05

Table 33: Per-pass performance for `OctTree`, ADT has 16 fields, SoA uses 9 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across -iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Si	Am/Sm	Ai/Am	Ai/Sm	Ai/Si
clearFlags	M	15/16	6%	0.0193 $\pm$ 0.00086	0.0192$\pm$0.00002	0.0245 $\pm$ 0.00002	0.0669 $\pm$ 0.00002	0.79 $\times$	0.99 $\times$	0.78 $\times$	0.29 $\times$
countActive	F	9/16	44%	0.0039 $\pm$ 0.00000	0.0054 $\pm$ 0.00000	0.0032$\pm$0.00000	0.0277 $\pm$ 0.00000	1.22$\times$	1.39$\times$	1.70$\times$	0.20 $\times$
countParticles	F	8/16	50%	0.0039 $\pm$ 0.00001	0.0052 $\pm$ 0.00000	0.0030$\pm$0.00000	0.0266 $\pm$ 0.00000	1.29$\times$	1.31$\times$	1.70$\times$	0.19 $\times$
scaleEnergy	M	16/16	0%	0.0192 $\pm$ 0.00002	0.0191$\pm$0.00003	0.0246 $\pm$ 0.00013	0.0700 $\pm$ 0.00098	0.78 $\times$	1.00 $\times$	0.78 $\times$	0.27 $\times$
sumEnergy	F	10/16	38%	0.0040 $\pm$ 0.00000	0.0054 $\pm$ 0.00000	0.0034$\pm$0.00000	0.0280 $\pm$ 0.00001	1.16$\times$	1.37$\times$	1.58$\times$	0.19 $\times$
sumMass	F	9/16	44%	0.0040 $\pm$ 0.00033	0.0054 $\pm$ 0.00047	0.0034$\pm$0.00028	0.0278 $\pm$ 0.00022	1.18$\times$	1.36$\times$	1.61$\times$	0.20 $\times$
Total				0.0542	0.0597	0.0622	0.2471	0.87 $\times$	1.10$\times$	0.96 $\times$	0.24 $\times$
Geomean								1.05 $\times$	1.22$\times$	1.28$\times$	0.22 $\times$

Table 34: Per-pass PAPI counters for `OctTree` (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
clearFlags	M	4.6e+07 /6.7e+07	1.0e+08 /2.6e+08	7.9e+04 /3.9e+05	2.4e+04 /5.9e+04	1.3e+03 /5.2e+03	9.2e+02 /2.0e+03	5.6e+05/ 2.4e+05
countActive	F	2.0e+07/ 1.6e+07	5.1e+07/ 4.9e+07	5.3e+04/ 4.5e+04	8.8e+01/ 3.0e+01	4.7e+02 /1.6e+03	8.8e+01/ 3.2e+01	6.1e+05/ 2.4e+03
countParticles	F	2.0e+07/ 1.5e+07	5.2e+07/ 5.2e+07	6.7e+04/ 2.9e+04	1.4e+02/ 3.5e+01	6.3e+02 /8.3e+02	1.4e+02/ 3.6e+01	6.2e+05/ 4.1e+02
scaleEnergy	M	4.6e+07 /6.9e+07	1.0e+08 /2.7e+08	7.8e+04 /4.4e+05	2.9e+04 /5.4e+04	1.7e+03 /5.6e+03	1.6e+03 /2.0e+03	5.6e+05/ 2.5e+05
sumEnergy	F	2.0e+07/ 1.7e+07	5.2e+07 /6.1e+07	2.6e+04/ 2.1e+04	9.0e+01/ 4.4e+01	4.0e+02 /1.4e+03	9.0e+01/ 4.5e+01	6.0e+05/ 4.1e+04
sumMass	F	2.0e+07/ 1.7e+07	5.2e+07 /6.1e+07	2.5e+04 /2.6e+04	9.6e+01/ 4.0e+01	4.1e+02 /1.5e+03	9.8e+01/ 4.8e+01	6.1e+05/ 4.2e+04
Total		1.7e+08/2.0e+08	4.1e+08/7.5e+08	3.3e+05/9.6e+05	5.3e+04/1.1e+05	4.9e+03/1.6e+04	2.9e+03/4.1e+03	3.6e+06/5.8e+05

Table 35: Per-pass PAPI counters for OctTree (Ai/Si). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Ai/Si)	INS (Ai/Si)	L1D (Ai/Si)	L1I (Ai/Si)	L2D (Ai/Si)	L2I (Ai/Si)	LLC (Ai/Si)
clearFlags	M	4.6e+07 /2.8e+08	1.3e+08 /6.5e+08	1.4e+05 /1.1e+06	2.5e+04 /1.5e+05	1.1e+03 /8.7e+03	8.2e+02 /3.3e+03	5.7e+05/ 2.4e+05
countActive	F	2.7e+07 /1.4e+08	8.9e+07 /2.5e+08	1.6e+05/ 7.7e+04	7.8e+01 /3.8e+02	3.7e+02 /4.4e+03	7.8e+01/ 6.5e+01	5.4e+05/ 3.7e+03
countParticles	F	2.6e+07 /1.4e+08	8.3e+07 /2.5e+08	4.3e+04/ 4.2e+04	8.3e+01 /4.0e+02	3.4e+02 /4.0e+03	8.3e+01/ 6.4e+01	5.7e+05/ 1.6e+03
scaleEnergy	M	4.6e+07 /3.0e+08	1.4e+08 /6.4e+08	1.3e+05 /1.1e+06	2.9e+04 /1.4e+05	9.2e+02 /7.8e+03	7.2e+02 /2.6e+03	5.5e+05/ 2.5e+05
sumEnergy	F	2.7e+07 /1.4e+08	9.1e+07 /2.5e+08	1.5e+04 /3.3e+05	7.4e+01 /3.2e+02	3.6e+02 /8.1e+03	7.4e+01 /2.0e+02	5.5e+05/ 4.7e+04
sumMass	F	2.7e+07 /1.4e+08	9.1e+07 /2.5e+08	1.4e+04 /3.3e+05	7.4e+01 /3.8e+02	3.5e+02 /8.3e+03	7.4e+01 /2.4e+02	5.6e+05/ 5.1e+04
Total		2.0e+08/1.1e+09	6.2e+08/2.3e+09	5.0e+05/3.1e+06	5.4e+04/2.9e+05	3.4e+03/4.1e+04	1.9e+03/6.4e+03	3.3e+06/5.9e+05

Table 36: Per-pass performance for PiecewiseFunctions, ADT has 8 fields, SoA uses 7 buffers (mutable + immutable cursors). Times are median per iteration (s); ± shows standard error of the mean across –iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup >1× means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Si	Am/Sm	Ai/Am	Ai/Sm	Ai/Si
add constant	M	8/8	0%	0.0150±0.00002	0.0169±0.00002	0.0171±0.00003	0.0554±0.00002	0.88×	1.13×	0.99×	0.31×
countSplit	F	3/8	62%	0.0026±0.00000	0.0067±0.00000	0.0013±0.00000	0.0338±0.00107	2.05×	2.61×	5.35×	0.20×
differentiate	M	8/8	0%	0.0151±0.00002	0.0172±0.00002	0.0170±0.00086	0.0561±0.00103	0.89×	1.13×	1.01×	0.31×
max degree	F	3/8	62%	0.0027±0.00000	0.0058±0.00000	0.0015±0.00019	0.0274±0.00000	1.84×	2.14×	3.93×	0.21×
square	M	8/8	0%	0.0149±0.00002	0.0167±0.00002	0.0169±0.00084	0.0562±0.00002	0.88×	1.12×	0.99×	0.30×
sum co-efficients	F	3/8	62%	0.0025±0.00020	0.0059±0.00050	0.0014±0.00027	0.0335±0.00037	1.76×	2.38×	4.20×	0.17×
sumError	F	3/8	62%	0.0024±0.00000	0.0058±0.00000	0.0014±0.00001	0.0315±0.00001	1.76×	2.39×	4.20×	0.18×
Total				0.0552	0.0750	0.0566	0.2939	0.98×	1.36×	1.33×	0.26×
Geomean								1.35×	1.73×	2.32×	0.23×

Table 37: Per-pass PAPI counters for PiecewiseFunctions (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
add constant	M	3.4e+07 /4.5e+07	7.4e+07 /1.7e+08	1.4e+05 /2.6e+05	1.9e+04 /2.8e+04	1.5e+03 /3.4e+03	1.6e+03 /3.9e+03	4.3e+05/ 1.8e+05
countSplit	F	1.3e+07/ 6.4e+06	3.5e+07 /3.9e+07	3.4e+05/ 1.4e+04	2.7e+02/ 5.2e+01	2.5e+03/ 6.2e+02	2.6e+02/ 5.3e+01	4.9e+05/ 6.6e+02
differentiate	M	3.5e+07 /4.5e+07	7.5e+07 /1.8e+08	1.1e+05 /2.2e+05	2.0e+04 /2.6e+04	1.5e+03 /2.9e+03	1.7e+03 /2.5e+03	4.8e+05/ 1.8e+05
max degree	F	1.4e+07/ 7.5e+06	3.4e+07 /3.7e+07	2.9e+05/ 4.5e+03	1.7e+02/ 5.5e+01	1.1e+03/ 6.9e+02	1.8e+02/ 5.5e+01	4.8e+05/ 7.5e+02
square	M	3.4e+07 /4.5e+07	7.4e+07 /1.7e+08	1.4e+05 /2.4e+05	2.0e+04 /2.8e+04	1.4e+03 /3.4e+03	1.4e+03 /4.1e+03	4.3e+05/ 1.8e+05
sum co-efficients	F	1.2e+07/ 7.0e+06	2.8e+07 /3.3e+07	3.5e+05/ 9.5e+03	2.2e+02/ 8.4e+01	3.2e+03/ 7.4e+02	2.2e+02/ 8.7e+01	4.9e+05/ 1.5e+03
sumError	F	1.2e+07/ 7.0e+06	2.8e+07 /3.3e+07	3.5e+05/ 8.8e+03	2.3e+02/ 7.1e+01	3.1e+03/ 6.7e+02	2.4e+02/ 7.1e+01	4.8e+05/ 6.0e+02
Total		1.6e+08/1.6e+08	3.5e+08/6.6e+08	1.7e+06/7.5e+05	6.0e+04/8.2e+04	1.4e+04/1.2e+04	5.6e+03/1.1e+04	3.3e+06/5.4e+05

Table 38: Per-pass PAPI counters for PiecewiseFunctions (Ai/Si). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Ai/Si)	INS (Ai/Si)	L1D (Ai/Si)	L1I (Ai/Si)	L2D (Ai/Si)	L2I (Ai/Si)	LLC (Ai/Si)
add constant	M	4.4e+07 /2.4e+08	1.2e+08 /4.4e+08	4.5e+04 /6.6e+05	2.6e+04 /3.6e+04	8.4e+02 /5.7e+03	1.0e+03 /4.1e+03	4.0e+05/ 1.9e+05
countSplit	F	3.4e+07 /1.7e+08	8.2e+07 /2.0e+08	5.8e+05/ 1.5e+05	5.6e+01 /2.6e+02	3.7e+02 /3.4e+03	5.5e+01 /9.1e+01	3.2e+05/ 1.2e+03
differentiate	M	4.5e+07 /2.4e+08	1.3e+08 /4.6e+08	4.1e+04 /6.9e+05	1.7e+04 /4.6e+04	7.6e+02 /7.1e+03	7.9e+02 /4.1e+03	4.4e+05/ 1.9e+05
max degree	F	2.9e+07 /1.4e+08	7.8e+07 /1.9e+08	2.7e+04 /1.6e+05	4.9e+01 /2.6e+02	2.9e+02 /3.6e+03	4.9e+01 /7.9e+01	3.7e+05/ 1.9e+03
square	M	4.4e+07 /2.4e+08	1.2e+08 /4.4e+08	4.9e+04 /6.4e+05	2.3e+04 /3.5e+04	9.8e+02 /5.9e+03	8.6e+02 /6.1e+03	4.0e+05/ 1.9e+05
sum co-efficients	F	3.0e+07 /1.7e+08	7.6e+07 /1.9e+08	2.6e+04 /1.6e+05	5.7e+01 /2.8e+02	2.9e+02 /3.7e+03	5.7e+01 /8.1e+01	3.8e+05/ 2.7e+03
sumError	F	2.9e+07 /1.6e+08	7.6e+07 /1.9e+08	2.5e+04 /1.6e+05	6.9e+01 /3.2e+02	2.7e+02 /3.5e+03	7.0e+01 /8.9e+01	3.6e+05/ 7.7e+02
Total		2.6e+08/1.4e+09	6.8e+08/2.1e+09	7.9e+05/2.6e+06	6.7e+04/1.2e+05	3.8e+03/3.3e+04	2.9e+03/1.5e+04	2.7e+06/5.7e+05

Table 39: Per-pass performance for TernaryTree, ADT has 5 fields, SoA uses 3 buffers (mutable + immutable cursors). Times are median per iteration (s); ± shows standard error of the mean across –iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup >1× means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Si	Am/Sm	Ai/Am	Ai/Sm	Ai/Si
add 1 tree	M	5/5	0%	0.0845±0.00007	0.0967±0.00016	0.0771±0.00121	0.3656±0.00211	1.10×	1.14×	1.25×	0.26×
sum tree	F	5/5	0%	0.0301±0.00006	0.0525±0.00001	0.0283±0.00012	0.2521±0.00115	1.06×	1.75×	1.86×	0.21×
Total				0.1145	0.1492	0.1054	0.6177	1.09×	1.30×	1.42×	0.24×
Geomean								1.08×	1.41×	1.53×	0.23×

Table 40: Per-pass PAPI counters for TernaryTree (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
add 1 tree	M	2.7e+08/ 2.2e+08	7.5e+08 /1.1e+09	3.7e+05 /9.9e+05	7.1e+04 /1.4e+05	4.3e+03 /1.1e+04	4.0e+03/ 3.6e+03	1.5e+06/ 1.2e+06
sum tree	F	1.5e+08/ 1.4e+08	4.8e+08 /5.1e+08	2.9e+04 /1.5e+05	8.3e+01 /2.6e+02	8.2e+02 /9.0e+03	8.3e+01 /2.6e+02	1.7e+06/ 1.2e+06
Total		4.3e+08/3.7e+08	1.2e+09/1.6e+09	4.0e+05/1.1e+06	7.2e+04/1.4e+05	5.1e+03/2.0e+04	4.0e+03/3.9e+03	3.2e+06/2.4e+06

Table 41: Per-pass PAPI counters for TernaryTree (Ai/Si). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Ai/Si)	INS (Ai/Si)	L1D (Ai/Si)	L1I (Ai/Si)	L2D (Ai/Si)	L2I (Ai/Si)	LLC (Ai/Si)
add 1 tree	M	3.4e+08 /1.7e+09	1.2e+09 /2.7e+09	1.4e+05 /2.5e+06	7.1e+04 /2.0e+05	2.7e+03 /4.6e+04	3.0e+03 /6.0e+03	1.4e+06/ 9.5e+05
sum tree	F	2.7e+08 /1.3e+09	8.3e+08 /1.3e+09	1.3e+05 /2.7e+06	1.7e+02 /6.8e+02	1.5e+03 /4.8e+04	1.7e+02 /2.2e+02	1.3e+06/ 8.7e+05
Total		6.1e+08/3.0e+09	2.0e+09/4.0e+09	2.6e+05/5.2e+06	7.1e+04/2.0e+05	4.2e+03/9.4e+04	3.1e+03/6.2e+03	2.7e+06/1.8e+06

Table 42: Per-pass performance for Trie, ADT has 10 fields, SoA uses 9 buffers (mutable + immutable cursors). Times are median per iteration (s); ± shows standard error of the mean across –iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup >1× means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Si	Am/Sm	Ai/Am	Ai/Sm	Ai/Si
clear trie flags	M	9/10	10%	0.0375±0.00007	0.0422±0.00005	0.0466±0.00007	0.1257±0.00021	0.81×	1.12×	0.91×	0.34×
count trie flags	F	3/10	70%	0.0062±0.00000	0.0127±0.00001	0.0028±0.00000	0.0508±0.00002	2.19×	2.05×	4.48×	0.25×
scale frequency	M	10/10	0%	0.0380±0.00017	0.0417±0.00006	0.0479±0.00120	0.1265±0.00007	0.79×	1.10×	0.87×	0.33×
sum count words	F	2/10	80%	0.0060±0.00000	0.0137±0.00001	0.0026±0.00000	0.0569±0.00127	2.35×	2.28×	5.35×	0.24×
sum frequency	F	3/10	70%	0.0062±0.00019	0.0125±0.00036	0.0028±0.00018	0.0547±0.00112	2.17×	2.03×	4.41×	0.23×
sum subtrees	F	3/10	70%	0.0062±0.00000	0.0125±0.00001	0.0028±0.00000	0.0547±0.00109	2.17×	2.03×	4.40×	0.23×
Total				0.1000	0.1354	0.1056	0.4692	0.95×	1.35×	1.28×	0.29×
Geomean								1.58×	1.70×	2.67×	0.27×

Table 43: Per-pass PAPI counters for `Trie` (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
clear trie flags	M	8.3e+07 /1.3e+08	1.4e+08 /4.3e+08	3.7e+05 /5.2e+05	4.4e+04 /6.9e+04	4.4e+03 /9.4e+03	1.6e+03 /1.4e+04	1.5e+06/ 5.0e+05
count trie flags	F	3.1e+07/ 1.4e+07	6.3e+07 /6.4e+07	7.6e+05/ 6.5e+03	5.7e+01/ 4.6e+01	8.9e+03/ 1.3e+03	5.7e+01/ 4.7e+01	1.9e+06/ 3.3e+04
scale frequency	M	8.3e+07 /1.3e+08	1.5e+08 /4.5e+08	3.4e+05 /6.8e+05	6.7e+04 /8.7e+04	4.1e+03 /1.0e+04	1.4e+03 /1.6e+04	1.5e+06/ 5.7e+05
sum count words	F	3.0e+07/ 1.3e+07	5.1e+07 /5.3e+07	8.2e+05/ 2.2e+04	8.5e+01/ 1.3e+01	2.3e+04/ 5.6e+02	8.5e+01/ 1.3e+01	1.8e+06/ 2.1e+03
sum frequency	F	3.1e+07/ 1.4e+07	5.7e+07 /5.8e+07	7.7e+05/ 6.1e+03	4.7e+01 /5.5e+01	1.8e+04/ 1.4e+03	4.7e+01 /5.6e+01	1.9e+06/ 3.4e+04
sum subtrees	F	3.1e+07/ 1.4e+07	5.7e+07 /5.8e+07	8.3e+05/ 6.5e+03	5.0e+01/ 4.9e+01	9.6e+03/ 1.4e+03	5.1e+01/ 4.9e+01	1.8e+06/ 3.2e+04
Total		2.9e+08/3.2e+08	5.2e+08/1.1e+09	3.9e+06/1.2e+06	1.1e+05/1.6e+05	6.8e+04/2.5e+04	3.2e+03/3.0e+04	1.0e+07/1.2e+06

Table 44: Per-pass PAPI counters for `Trie` (Ai/Si). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Ai/Si)	INS (Ai/Si)	L1D (Ai/Si)	L1I (Ai/Si)	L2D (Ai/Si)	L2I (Ai/Si)	LLC (Ai/Si)
clear trie flags	M	1.0e+08 /5.3e+08	2.3e+08 /1.1e+09	1.3e+05 /2.0e+06	4.4e+04 /2.2e+05	2.3e+03 /1.3e+04	2.0e+03 /2.2e+04	1.4e+06/ 5.0e+05
count trie flags	F	6.4e+07 /2.6e+08	1.6e+08 /4.5e+08	9.7e+04 /3.1e+05	6.1e+01 /4.6e+02	6.0e+02 /7.9e+03	6.1e+01 /1.0e+02	1.3e+06/ 5.1e+04
scale frequency	M	1.0e+08 /5.3e+08	2.4e+08 /1.1e+09	1.1e+05 /2.4e+06	5.2e+04 /8.3e+04	2.2e+03 /1.6e+04	2.0e+03 /2.4e+04	1.4e+06/ 5.7e+05
sum count words	F	6.9e+07 /2.9e+08	1.4e+08 /4.2e+08	1.1e+06/ 6.7e+04	4.7e+01 /5.8e+02	1.3e+03 /4.3e+03	4.7e+01 /7.8e+01	1.3e+06/ 4.9e+03
sum frequency	F	6.4e+07 /2.8e+08	1.5e+08 /4.2e+08	1.2e+05 /3.1e+05	4.7e+01 /5.1e+02	6.6e+02 /8.0e+03	4.6e+01 /1.2e+02	1.4e+06/ 5.2e+04
sum subtrees	F	6.3e+07 /2.8e+08	1.5e+08 /4.2e+08	1.1e+05 /3.0e+05	6.3e+01 /3.7e+02	6.3e+02 /7.9e+03	6.3e+01 /1.1e+02	1.3e+06/ 5.1e+04
Total		4.7e+08/2.2e+09	1.1e+09/4.0e+09	1.7e+06/5.4e+06	9.6e+04/3.0e+05	7.7e+03/5.7e+04	4.2e+03/4.6e+04	8.1e+06/1.2e+06